

Mathematics for Aerospace Engineers

Chapter 3: Calculus of functions of multiple variables

School of Mathematical Sciences

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**University of
Nottingham**

UK | CHINA | MALAYSIA

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Calculus of functions of multiple variables

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3.1 Introduction

Often, physical quantities depend upon more than one variable.

- In 1-D fluid flow, the fluid velocity $\mathbf{u}$ might depend upon space x and time t . We would then write $\mathbf{u}(x, t)$ to make this explicit.
- In 2-D plate vibrations, the vertical plate deflection Z might depend upon the coordinates (x, y) of a point in the plate, so $Z = Z(x, y)$.
- Notation: If z is a function of x and y we write $z = f(x, y)$ where f is the *dependent* variable and x and y are the *independent* variables.

3.2 Surfaces

The graph of a function of one variable $y = f(x)$ gives us a 1-D curve in the (x, y) plane.

- For a circle of radius a , $x^2 + y^2 = a^2$. So $y = f(x) = \pm (a^2 - x^2)^{\frac{1}{2}}$ and this equation gives $f(x)$ for the circular curve.
- For a sphere of radius a , $x^2 + y^2 + z^2 = a^2$. We can then write $z = f(x, y) = \pm (a^2 - x^2 - y^2)^{\frac{1}{2}}$.
This is the equation of the 2-D spherical surface.

Generally, just as $y = f(x)$ represents a 1-D curve, $z = f(x, y)$ represents a 2-D surface.

3.3 Partial Derivatives

Suppose we fix the value of y (or x) in the equation of the surface $z = f(x, y)$. Then z depends only upon x (or y) and we get back to the equation of a curve.

This is just the same as looking at cross-sections through the surface. This curve has a gradient given by the derivative of z with respect to x (or y).

So it makes sense to consider the derivatives of $f(x, y)$ with respect to x holding y fixed or with respect to y holding x fixed.

These are called **partial derivatives**.

3.3.1 Notation for Partial Derivatives

The **first-order partial derivative** of $f(x, y)$ with respect to x (holding y fixed) is written

$$\frac{\partial f}{\partial x} \quad \text{or} \quad f_x$$

Similarly, the partial derivative of $f(x, y)$ with respect to y (holding x fixed) is

$$\frac{\partial f}{\partial y} \quad \text{or} \quad f_y$$

Higher-order partial derivatives are possible - we return to this later.

3.3.2 Calculating Partial Derivatives

We now calculate a partial derivative from first principles, and use this to motivate an easier way of doing it.

Consider $f(x, y) = xy$

Recalling the definition of a derivative, we see that

$$\begin{aligned}\frac{\partial f}{\partial x} &= \lim_{h \rightarrow 0} \frac{f(x+h, y) - f(x, y)}{h} \\ &= \lim_{h \rightarrow 0} \frac{(x+h)y - xy}{h} \\ &= \lim_{h \rightarrow 0} \frac{xy + hy - xy}{h} = y.\end{aligned}$$

Hence, in this case $\frac{\partial f}{\partial x} = y$.

The partial derivative with respect to x treats y as though it were a constant (and vice versa).

Hence, if $f(x, y) = xy$ then the partial derivative of f with respect to x is simply the derivative of xy with y remaining a constant.

$$\text{i.e. } \frac{\partial f}{\partial x} = y$$

and the answer is immediate. Similarly,

$$\frac{\partial f}{\partial y} = x.$$

Example 3.3.1

Find the first order partial derivatives of $f(x, y) = \sin(xy) + \frac{1}{2}y^2$.

Solution:

$$\frac{\partial f}{\partial x} = y \cos(xy) + 0 = y \cos(xy).$$

$$\text{Similarly, } \frac{\partial f}{\partial y} = x \cos(xy) + y.$$

Example 3.3.2

Find the first order partial derivatives of $f(x, y) = x^4 + 3x^2y + y^5$.

Solution:

$$\frac{\partial f}{\partial x} = 4x^3 + 6xy \quad \text{and} \quad \frac{\partial f}{\partial y} = 3x^2 + 5y^4.$$

Example 3.3.3

Find the first order partial derivatives of $f(x, y) = \frac{x}{y}$.

Solution: $\frac{\partial f}{\partial x} = \frac{1}{y}, \quad \frac{\partial f}{\partial y} = -\frac{x}{y^2}.$

Notice that in this case

$$x \frac{\partial f}{\partial x} + y \frac{\partial f}{\partial y} = \frac{x}{y} + y \left(-\frac{x}{y^2} \right) = \frac{x}{y} - \frac{x}{y} = 0.$$

Hence, the first-order partial derivatives of $f(x, y) = x/y$ satisfy

$$x \frac{\partial f}{\partial x} + y \frac{\partial f}{\partial y} = 0.$$

(This is an example of a 'partial differential equation', which you will study in a Second Year module.)

Example 3.3.4

Find the first order partial derivatives of $f(x, y) = \frac{1}{x + y^2}$.

Solution: $\frac{\partial f}{\partial x} = -\frac{1}{(x + y^2)^2}$ and

$$\frac{\partial f}{\partial y} = \frac{\partial}{\partial y} (x + y^2)^{-1} = -\frac{1}{(x + y^2)^2} \frac{\partial}{\partial y} (x + y^2) = -\frac{2y}{(x + y^2)^2}.$$

Example 3.3.5

Find the first order partial derivatives of $f(x, y) = (2x + 3y)^4$.

Solution:

$$\begin{aligned}\frac{\partial f}{\partial x} &= \frac{\partial}{\partial x}(2x + 3y)^4 \\ &= 4(2x + 3y)^3 \frac{\partial}{\partial x}(2x + 3y) = 8(2x + 3y)^3.\end{aligned}$$

Exercise: show that

$$\frac{\partial f}{\partial y} = 12(2x + 3y)^3.$$

3.3.3 General Points

$f(x, y)$ has two first-order partial derivatives,

$$\frac{\partial f}{\partial x} \quad \text{and} \quad \frac{\partial f}{\partial y}.$$

Also,

$$\frac{\partial g(y)}{\partial x} = 0, \quad \frac{\partial h(x)}{\partial y} = 0$$

and

$$\frac{\partial h(x)}{\partial x} = \frac{dh(x)}{dx}; \quad \frac{\partial g(y)}{\partial y} = \frac{dg(y)}{dy}.$$

3.4 Second-Order Partial Derivatives

If $f(x, y) = x^2y^3$, then

$$\frac{\partial f}{\partial x} = 2xy^3 \quad \text{and} \quad \frac{\partial f}{\partial y} = 3x^2y^2.$$

We can differentiate these again:

$$\begin{aligned} \frac{\partial}{\partial x} \left(\frac{\partial f}{\partial x} \right) &= \frac{\partial}{\partial x} (2xy^3) = 2y^3, \\ \frac{\partial}{\partial y} \left(\frac{\partial f}{\partial y} \right) &= \frac{\partial}{\partial y} (3x^2y^2) = 6x^2y. \\ \text{Also, } \frac{\partial}{\partial y} \left(\frac{\partial f}{\partial x} \right) &= \frac{\partial}{\partial y} (2xy^3) = 6xy^2, \\ \frac{\partial}{\partial x} \left(\frac{\partial f}{\partial y} \right) &= \frac{\partial}{\partial x} (3x^2y^2) = 6xy^2. \end{aligned}$$

Notice that the last two expressions are equal.

3.4.1 Notation for Second-Order Partial Derivatives

There are *three* **second-order partial derivatives**:

$$\frac{\partial}{\partial x} \left(\frac{\partial f}{\partial x} \right) = \frac{\partial^2 f}{\partial x^2} = f_{xx}, \quad \frac{\partial}{\partial y} \left(\frac{\partial f}{\partial y} \right) = \frac{\partial^2 f}{\partial y^2} = f_{yy}$$

and

$$\frac{\partial}{\partial x} \left(\frac{\partial f}{\partial y} \right) = \frac{\partial}{\partial y} \left(\frac{\partial f}{\partial x} \right) = \frac{\partial^2 f}{\partial x \partial y} = \frac{\partial^2 f}{\partial y \partial x} = f_{xy} = f_{yx}.$$

The last derivative is called a **mixed derivative** and it doesn't depend upon the order of differentiation (x first then y, or y first then x are equally good.)

Example 3.4.1

Find the first and second order derivatives of $f(x, y) = y \ln x$.

Solution: $\frac{\partial f}{\partial x} = \frac{y}{x}, \quad \frac{\partial f}{\partial y} = \ln x.$

$$\frac{\partial^2 f}{\partial x^2} = \frac{\partial}{\partial x} \left(\frac{\partial f}{\partial x} \right) = -\frac{y}{x^2}.$$

$$\frac{\partial^2 f}{\partial y^2} = \frac{\partial}{\partial y} \left(\frac{\partial f}{\partial y} \right) = \frac{\partial(\ln x)}{\partial y} = 0.$$

$$\frac{\partial^2 f}{\partial x \partial y} = \frac{\partial}{\partial x} \left(\frac{\partial f}{\partial y} \right) = \frac{1}{x}$$

$$\frac{\partial^2 f}{\partial y \partial x} = \frac{\partial}{\partial y} \left(\frac{\partial f}{\partial x} \right) = \frac{1}{x}.$$

Example 3.4.2

Find the first and second order partial derivatives of

$$f(x, y) = \cos(x + y^2).$$

Solution:
$$\frac{\partial f}{\partial x} = -\sin(x + y^2)$$

$$\frac{\partial f}{\partial y} = -\sin(x + y^2) \frac{\partial}{\partial y} (x + y^2) = -2y \sin(x + y^2).$$

$$\frac{\partial^2 f}{\partial x^2} = \frac{\partial}{\partial x} \left(-\sin(x + y^2) \right) = -\cos(x + y^2)$$

$$\begin{aligned} \frac{\partial^2 f}{\partial y^2} &= \frac{\partial}{\partial y} \left(-2y \sin(x + y^2) \right) \\ &= \sin(x + y^2) \frac{\partial}{\partial y} (-2y) + (-2y) \frac{\partial}{\partial y} \sin(x + y^2) \\ &= -2 \sin(x + y^2) + (-2y) \left(2y \cos(x + y^2) \right). \end{aligned}$$

$$\frac{\partial^2 f}{\partial y^2} = -2 \sin(x + y^2) - 4y^2 \cos(x + y^2).$$

$$\begin{aligned}\frac{\partial^2 f}{\partial x \partial y} &= \frac{\partial}{\partial x} \left(\frac{\partial f}{\partial y} \right) \\ &= \frac{\partial}{\partial x} \left(-2y \sin(x + y^2) \right) = -2y \cos(x + y^2).\end{aligned}$$

Exercise: Check the last result by computing the mixed derivative in the opposite order (i.e. differentiate with respect to x first, then y to find

$$\frac{\partial^2 f}{\partial y \partial x} = \frac{\partial}{\partial y} \left(\frac{\partial f}{\partial x} \right).$$

Example 3.4.3

Given $f(x, y) = \ln\left((x^2 + y^2)^{\frac{1}{2}}\right)$, find f_x, f_y, f_{xx}, f_{yy} and show $f_{xx} + f_{yy} = 0$.

Solution: $\frac{\partial f}{\partial x} = \frac{\partial}{\partial x} \ln\left((x^2 + y^2)^{\frac{1}{2}}\right) = \frac{1}{(x^2 + y^2)^{\frac{1}{2}}} \frac{\partial}{\partial x} (x^2 + y^2)^{\frac{1}{2}}$

using the 'function of a function' rule for differentiation. Hence,

$$\frac{\partial f}{\partial x} = \frac{1}{(x^2 + y^2)^{\frac{1}{2}}} \times \frac{1}{2} \times \frac{1}{(x^2 + y^2)^{\frac{1}{2}}} \times \frac{\partial}{\partial x} (x^2 + y^2)$$

$$\text{i.e. } \frac{\partial f}{\partial x} = \frac{x}{x^2 + y^2}.$$

Similarly (exercise),

$$\frac{\partial f}{\partial y} = \frac{y}{x^2 + y^2}.$$

$$\begin{aligned}\frac{\partial^2 f}{\partial x^2} &= \frac{\partial}{\partial x} \left(\frac{x}{x^2+y^2} \right) = \left(\frac{1}{x^2+y^2} \right) \frac{\partial}{\partial x}(x) + x \frac{\partial}{\partial x} \left(\frac{1}{x^2+y^2} \right) \\ &= \frac{1}{x^2+y^2} + x \frac{(-2x)}{(x^2+y^2)^2} = \frac{y^2-x^2}{(x^2+y^2)^2},\end{aligned}$$

after some simplification (exercise).

Also, another calculation (exercise) gives

$$\frac{\partial^2 f}{\partial y^2} = \frac{x^2-y^2}{(x^2+y^2)^2}.$$

Hence, the function f satisfies the ‘partial differential equation’

$$\frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} = 0.$$

3.5 Partial Derivatives of Functions of Three Variables

Suppose we have a function depending on three variables, such as $f(x, y, z)$.

We can still calculate partial derivatives exactly as before, by treating the other variables as constants when performing the differentiation.

Example: $f(x, y, z) = xy^2z^3$

$$\frac{\partial f}{\partial x} = y^2z^3, \quad \frac{\partial f}{\partial y} = 2xyz^3, \quad \frac{\partial f}{\partial z} = 3xy^2z^2.$$

Example 3.5.1

Find the first order partial derivatives of $f(x, y, z) = (x + 2y - z)^3$.

Solution: $\frac{\partial f}{\partial x} = 3(x + 2y - z)^2, \quad \frac{\partial f}{\partial y} = 6(x + 2y - z)^2,$

$$\frac{\partial f}{\partial z} = -3(x + 2y - z)^2.$$

Example 3.5.2

Find the first order partial derivatives of $f(x, y, z) = z^{\frac{1}{2}} e^{xy}$.

Solution: $\frac{\partial f}{\partial x} = yz^{1/2} e^{xy}, \quad \frac{\partial f}{\partial y} = xz^{1/2} e^{xy}, \quad \frac{\partial f}{\partial z} = \frac{e^{xy}}{2z^{1/2}}.$

3.6 Stationary Points for Functions of Two Variables

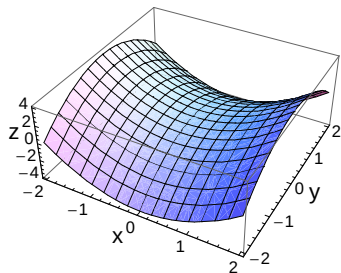
Surfaces $z = f(x, y)$ can have local **maxima** and **minima**, just as curves do. For example, the sphere $z = \sqrt{a^2 - x^2 - y^2}$ has a local maximum when $x = y = 0$, where the surface is locally flat.

There is one other possibility too - analogous to the 'inflection' stationary point for some curves. Imagine a surface of the shape of a saddle on a horse back. The middle of the surface is locally flat, and so we expect it to be a stationary point.

If we pass through this point along a path from the horses head to its tail, then it appears to be a minimum.

But if we take a path around the girth of the horse, it appears to be a maximum.

It is called a **saddle point**.



3.6.1 Conditions for Stationary Points

There are three types of stationary (or critical) points on surfaces: **maxima**, **minima** and **saddle points**.

We now show how to locate and classify them.

The following results can be proved from a detailed analysis of the Taylor series expansion; this proof is not required for this module.

Step One

For any stationary point of the function $f(x, y)$, we must have

$$\frac{\partial f}{\partial x} = 0 \quad \text{AND} \quad \frac{\partial f}{\partial y} = 0.$$

So we calculate these derivatives and find which values of x and y make them zero.

Step Two

We calculate the three second-order partial derivatives at each of these points, and then calculate the value of the '*Hessian*' Δ , where

$$\Delta = \begin{vmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{vmatrix} = \left(\frac{\partial^2 f}{\partial x^2} \right) \left(\frac{\partial^2 f}{\partial y^2} \right) - \left(\frac{\partial^2 f}{\partial x \partial y} \right)^2.$$

Step Three

- If $\Delta > 0$ **AND** $\frac{\partial^2 f}{\partial x^2} > 0$ (or $\frac{\partial^2 f}{\partial y^2} > 0$)

then the stationary point is a local **minimum**.

- If $\Delta > 0$ **AND** $\frac{\partial^2 f}{\partial x^2} < 0$ (or $\frac{\partial^2 f}{\partial y^2} < 0$)

then the stationary point is a local **maximum**.

- If $\Delta < 0$ then the stationary point is a **saddle point**.
- If $\Delta = 0$ the procedure is inconclusive.

Example 3.6.1

Verify that $(0, 0)$ is a stationary point of $f(x, y) = 5 - x^2 - y^2$, and determine its classification (as a maximum, minimum or saddle point).

Solution. $\frac{\partial f}{\partial x} = -2x, \quad \frac{\partial f}{\partial y} = -2y.$

Both derivatives vanish when $x = 0$ and $y = 0$.

The only stationary point for this function is the origin $(0, 0)$.

$$\text{Also, } \frac{\partial^2 f}{\partial x^2} = -2 = \frac{\partial^2 f}{\partial y^2}, \quad \frac{\partial^2 f}{\partial x \partial y} = 0,$$

$$\text{and so } \Delta = \left(\frac{\partial^2 f}{\partial x^2} \right) \left(\frac{\partial^2 f}{\partial y^2} \right) - \left(\frac{\partial^2 f}{\partial x \partial y} \right)^2 = (-2)(-2) - 0 = 4 > 0.$$

Further, $\frac{\partial^2 f}{\partial x^2} = -2 < 0$ and so $(0, 0)$ is a local maximum.

Example 3.6.2

Verify that $(0, 0)$ is a critical point of $f(x, y) = x^2 + y^2$, and determine its classification.

Solution. $\frac{\partial f}{\partial x} = 2x$, $\frac{\partial f}{\partial y} = 2y$. Both derivatives vanish when $x = 0$ and $y = 0$. The only stationary point for this function is the origin $(0, 0)$.

$$\text{Also, } \frac{\partial^2 f}{\partial x^2} = 2 = \frac{\partial^2 f}{\partial y^2}, \quad \frac{\partial^2 f}{\partial x \partial y} = 0,$$

$$\text{and so } \Delta = \left(\frac{\partial^2 f}{\partial x^2} \right) \left(\frac{\partial^2 f}{\partial y^2} \right) - \left(\frac{\partial^2 f}{\partial x \partial y} \right)^2 = (2)(2) - 0 = 4 > 0.$$

Further, $\frac{\partial^2 f}{\partial x^2} = 2 > 0$ and so $(0, 0)$ is a local minimum.

Example 3.6.3

Verify that $(0, 0)$ is a critical point of $f(x, y) = x^2 - y^2$, and determine its classification.

Solution. $\frac{\partial f}{\partial x} = 2x, \quad \frac{\partial f}{\partial y} = -2y.$

Both derivatives vanish when $x = 0$ and $y = 0$.

The only stationary point for this function is the origin $(0, 0)$.

$$\text{Also, } \frac{\partial^2 f}{\partial x^2} = 2, \quad \frac{\partial^2 f}{\partial y^2} = -2, \quad \frac{\partial^2 f}{\partial x \partial y} = 0,$$

$$\text{and so } \Delta = \left(\frac{\partial^2 f}{\partial x^2} \right) \left(\frac{\partial^2 f}{\partial y^2} \right) - \left(\frac{\partial^2 f}{\partial x \partial y} \right)^2 = (2)(-2) - 0 = -4 < 0.$$

Hence, $(0, 0)$ is a saddle point.

Example 3.6.4

Locate and classify the stationary points of the function

$$f(x, y) = \frac{1}{3}x^3 - x + \frac{1}{2}y^2 + 2y.$$

Solution. $\frac{\partial f}{\partial x} = x^2 - 1,$ $\frac{\partial f}{\partial y} = y + 2,$
and so at a stationary point,

$$x^2 - 1 = 0 \quad \text{and} \quad y + 2 = 0.$$

that is, $x = 1$ or -1 , and $y = -2$.

The stationary points are at $(-1, -2)$ and $(1, -2)$.

$$\frac{\partial^2 f}{\partial x^2} = 2x, \quad \frac{\partial^2 f}{\partial y^2} = 1, \quad \frac{\partial^2 f}{\partial x \partial y} = 0, \quad \Delta = 2x.$$

We now consider each point in turn.

Point 1: at $(-1, -2)$, $\Delta = (2x).1 - 0 = -2 < 0$.
Hence, $(-1, -2)$ is a saddle point.

Point 2: at $(1, -2)$, $\Delta = (2x).1 - 0 = 2 > 0$.

Also, $\frac{\partial^2 f}{\partial x^2} = 2x = 2 > 0$, hence, $(1, -2)$ is a local minimum.

Example 3.6.5

Locate and classify the stationary points of

$$f(x, y) = 3x^2 - 2xy + y^2 - 8y.$$

Solution. $\frac{\partial f}{\partial x} = 6x - 2y, \quad \frac{\partial f}{\partial y} = -2x + 2y - 8;$

$$\frac{\partial^2 f}{\partial x^2} = 6, \quad \frac{\partial^2 f}{\partial x \partial y} = -2, \quad \frac{\partial^2 f}{\partial y^2} = 2.$$

Hence, $\Delta = 6 \times 2 - 4 = 8 > 0$, and $\frac{\partial^2 f}{\partial x^2} = 6 > 0$,
so any stationary point (if there is one) must be a minimum.
At the stationary point, we require

$$6x - 2y = 0 \quad \text{and} \quad -2x + 2y - 8 = 0.$$

Solving these simultaneous equations yields that the minimum occurs at the point (2, 6).

Example 3.6.6

Locate and classify the stationary points of

$$f(x, y) = x^3 + y^3 - 3xy^2 + 2x.$$

Solution $\frac{\partial f}{\partial x} = 3x^2 - 3y^2 + 2; \quad \frac{\partial f}{\partial y} = 3y^2 - 6xy.$

Hence, at a stationary point,

$$3x^2 - 3y^2 + 2 = 0 \quad \text{and} \quad 3y^2 - 6xy = 0.$$

The second equation gives $y(y - 2x) = 0$.

The only possibilities are $y = 0$ and $y = 2x$.

Substituting $y = 0$ into the first equation gives $3x^2 + 2 = 0$, which has no real solutions for x .

Hence, the only possibility is $y = 2x$.

If $y = 2x$, then substituting into the first equation gives

$$3x^2 - 3(2x)^2 + 2 = 0,$$

which simplifies to $9x^2 = 2$,

with solutions

$$x = +\frac{\sqrt{2}}{3} \quad \text{and} \quad x = -\frac{\sqrt{2}}{3}.$$

Since $y = 2x$, the stationary points are $(\frac{\sqrt{2}}{3}, \frac{2\sqrt{2}}{3})$ and $(-\frac{\sqrt{2}}{3}, -\frac{2\sqrt{2}}{3})$.

To classify them, we calculate that

$$\frac{\partial^2 f}{\partial x^2} = 6x; \quad \frac{\partial^2 f}{\partial x \partial y} = -6y; \quad \frac{\partial^2 f}{\partial y^2} = 6y - 6x.$$

$$\begin{aligned} \text{Hence, } \Delta &= \left(\frac{\partial^2 f}{\partial x^2} \right) \left(\frac{\partial^2 f}{\partial y^2} \right) - \left(\frac{\partial^2 f}{\partial x \partial y} \right)^2 \\ &= 6x(6y - 6x) - 36y^2 = 36(xy - x^2 - y^2). \end{aligned}$$

$$\text{At } \left(\frac{\sqrt{2}}{3}, \frac{2\sqrt{2}}{3} \right), \quad \Delta = 36 \left(\frac{4}{9} - \frac{2}{9} - \frac{8}{9} \right) = -24 < 0.$$

Hence, this point is a saddle point.

$$\text{At } \left(-\frac{\sqrt{2}}{3}, -\frac{2\sqrt{2}}{3} \right), \quad \Delta = 36 \left(\frac{4}{9} - \frac{2}{9} - \frac{8}{9} \right) = -24 < 0.$$

Hence, this point is also a saddle point.

Example 3.6.7

Calculate the dimensions of an open rectangular box of volume 4m^3 and with minimum surface area.

Solution. Suppose the box has sides of length x , y and z . Then its volume is $V = xyz$ and its surface area is $A = 2zy + 2zx + xy$. Note that the box is open, so one face is missing.

$$V = 4 \quad \Rightarrow \quad z = \frac{4}{xy}. \quad \text{Hence} \quad A = \frac{8}{x} + \frac{8}{y} + xy,$$

$$\text{To minimise } A: \quad \frac{\partial A}{\partial x} = -\frac{8}{x^2} + y; \quad \frac{\partial A}{\partial y} = -\frac{8}{y^2} + x.$$

$$\text{So} \quad -\frac{8}{x^2} + y = 0 \quad \text{and} \quad -\frac{8}{y^2} + x = 0.$$

$$\text{Hence} \quad \frac{x^2 y}{y^2 x} = \frac{8}{8} = 1 \quad \Rightarrow \quad \frac{x}{y} = 1, \quad \text{or} \quad x = y.$$

$$x^2 y = 8 \quad \Rightarrow \quad x^3 = 8, \quad \text{so } x = y = 2. \quad \text{Finally, } xyz = 4 \Rightarrow z = 1.$$

N.B. We should really check that the surface area is *minimised*.

$$\frac{\partial^2 A}{\partial x^2} = \frac{16}{x^3} = 2 > 0.$$

Also,

$$\frac{\partial^2 A}{\partial y^2} = 2 \quad \text{and} \quad \frac{\partial^2 A}{\partial x \partial y} = 1$$

so that

$$\Delta = \left(\frac{\partial^2 A}{\partial x^2} \right) \left(\frac{\partial^2 A}{\partial y^2} \right) - \left(\frac{\partial^2 A}{\partial x \partial y} \right)^2 = 4 - 1 = 3.$$

i.e. $\Delta > 0$.

Hence, all of the conditions for this stationary point to be a minimum are satisfied.

3.7 Total Derivatives

Suppose $f(x, y)$ is given, and each of x and y depend upon another variable t , say.

$$x = x(t), \quad y = y(t).$$

Then $g(t) = f(x(t), y(t))$, say.

Question. Can we calculate $g'(t)$?

Example 3.7.1

If $f(x, y) = x^2 + y^3$, $x(t) = \sqrt{t}$, $y(t) = t^2$, find $g'(t)$.

One way is to substitute:

$$g(t) = f(x(t), y(t)) = (\sqrt{t})^2 + (t^2)^3 = t + t^6.$$

$$\text{Hence } g'(t) = 1 + 6t^5.$$

Alternative method - use the Chain Rule

$$\boxed{\frac{dg}{dt} = \frac{d}{dt}f(x(t), y(t)) = \frac{\partial f}{\partial x} \frac{dx}{dt} + \frac{\partial f}{\partial y} \frac{dy}{dt}}.$$

$$\frac{\partial f}{\partial x} = 2x, \quad \frac{\partial f}{\partial y} = 3y^2;$$

$$\frac{dx}{dt} = \frac{1}{2t^{\frac{1}{2}}}, \quad \frac{dy}{dt} = 2t.$$

$$\begin{aligned} \text{Hence } \frac{dg}{dt} &= \frac{d}{dt}f(x(t), y(t)) = 2x \frac{1}{2t^{\frac{1}{2}}} + 3y^2 2t \\ &= 2t^{\frac{1}{2}} \frac{1}{2t^{\frac{1}{2}}} + 3t^4 2t = 1 + 6t^5, \quad \text{as before.} \end{aligned}$$

Example 3.7.2

Using the chain rule, find $\frac{dg}{dt}$ where $g(t) = f(x(t), y(t))$,

$$f(x, y) = x \cos y \text{ and } x(t) = t^2, \quad y(t) = 2t.$$

Solution: $g(t) = f(x(t), y(t)) = t^2 \cos(2t)$

$$\begin{aligned} \text{and so } g'(t) &= \frac{\partial f}{\partial x} \frac{dx}{dt} + \frac{\partial f}{\partial y} \frac{dy}{dt} = \cos(y)2t + (-x \sin y)2 \\ &= 2t \cos(2t) - 2t^2 \sin(2t). \end{aligned}$$

(as can be confirmed by direct differentiation of $t^2 \cos(2t)$).

Example 3.7.3

Suppose the coordinates of a particle at time t satisfy

$$x(t) = a \cos(\omega t), \quad y(t) = a \sin(\omega t)$$

and that we consider $f(x, y) = x^2 + y^2$.

Show that $f(x, y)$ is constant along the line $x(t), y(t)$.

Solution: Hence $\frac{d}{dt}f(x(t), y(t)) = \frac{\partial f}{\partial x} \frac{dx}{dt} + \frac{\partial f}{\partial y} \frac{dy}{dt}$

$$= (2x)(-a\omega \sin(\omega t)) + (2y)(a\omega \cos(\omega t))$$

$$= -2a^2\omega \sin(\omega t) \cos(\omega t) + 2a^2\omega \sin(\omega t) \cos(\omega t) = 0.$$

Hence f is constant.

At $t = 0$, $x = a$ and $y = 0$. Hence at $t = 0$, $f = a^2$.

Since f is constant along the line $x(t), y(t)$, $f = a^2$ for all t .

3.8 Change of variables

3.8.1 Change of variables in two dimensions

Suppose $f(x, y)$ is given and that each of x and y depend upon two new variables u and v

$$x = x(u, v), \quad y = y(u, v).$$

Then if we substitute for x and y in terms of u and v into f we get a new function g depending only upon u and v .

$$\text{i.e. } g(u, v) = f(x(u, v), y(u, v)).$$

Example 3.8.1

Plane polar coordinates (using $u = r$, $v = \theta$ implies)

$$x(u, v) = x(r, \theta) = r \cos \theta, \quad \text{and} \quad y(u, v) = y(r, \theta) = r \sin \theta.$$

Then if $f(x, y) = x^2 + y^2 + x$,

$$g(r, \theta) = f(x(r, \theta), y(r, \theta)) = (r \cos \theta)^2 + (r \sin \theta)^2 + r \cos \theta = r^2 + r \cos \theta.$$

A natural question to ask is if we can calculate the derivatives of g with respect to r and θ without having to do this substitution, ie. can we generalise the chain rule to cope with functions of two variables?

The answer is **yes**: if $x = x(u, v)$ and $y = y(u, v)$ and $f(x, y)$ is given, then the derivatives of the function

$$g(u, v) = f(x(u, v), y(u, v))$$

with respect to u and v are given by

$$\frac{\partial g}{\partial u} = \frac{\partial f}{\partial x} \frac{\partial x}{\partial u} + \frac{\partial f}{\partial y} \frac{\partial y}{\partial u} \quad \text{and} \quad \frac{\partial g}{\partial v} = \frac{\partial f}{\partial x} \frac{\partial x}{\partial v} + \frac{\partial f}{\partial y} \frac{\partial y}{\partial v}.$$

3.8.2 Simplification

NOTE: If $x = x(t)$ and $y = y(t)$, then $u = t$, and there is no dependence upon v . Then

$$\frac{\partial x}{\partial v} = 0, \quad \frac{\partial y}{\partial v} = 0, \quad \text{and} \quad \frac{\partial x}{\partial u} = \frac{dx}{dt}, \quad \frac{\partial y}{\partial u} = \frac{dy}{dt}.$$

We can also write $g(t) = f(x(u, v), y(u, v))$ and the change of variables formula now gives us

$$\frac{dg}{dt} = \frac{\partial f}{\partial x} \frac{dx}{dt} + \frac{\partial f}{\partial y} \frac{dy}{dt}.$$

Hence, we have re-created the chain rule given in §7.

3.8.3 Change of variables in three dimensions

Suppose $f = f(x, y, z)$ is given and that each of x, y, z depend upon three new variables, u, v, w , by

$$x = x(u, v, w), \quad y = y(u, v, w), \quad z = z(u, v, w)$$

then if we substitute for x, y, z in terms of u, v, w we obtain a new function depending upon (u, v, w) ,

$$g(u, v, w) = f(x(u, v, w), y(u, v, w), z(u, v, w)),$$

and the rate of change of g w.r.t. u is governed by the chain rule

$$g_u = \frac{\partial g}{\partial u} = \frac{\partial f}{\partial x} \frac{\partial x}{\partial u} + \frac{\partial f}{\partial y} \frac{\partial y}{\partial u} + \frac{\partial f}{\partial z} \frac{\partial z}{\partial u},$$

and similarly for g_v, g_w .

Example 3.8.2

Given $f(x, y) = x^2 + y^2 + x$, find g_r, g_θ , where

$$g(r, \theta) = f(x(r, \theta), y(r, \theta)).$$

Solution. For plane polar coordinates (r, θ) , we have

$$x = r \cos \theta, \quad y = r \sin \theta, \quad \text{so that} \quad g(r, \theta) = r^2 + r \cos \theta.$$

Then

$$\begin{aligned} \frac{\partial g}{\partial r} &= \frac{\partial f}{\partial x} \frac{\partial x}{\partial r} + \frac{\partial f}{\partial y} \frac{\partial y}{\partial r}, \\ &= (2x + 1) \cos \theta + (2y) \sin \theta, \\ &= (2r \cos \theta + 1) \cos \theta + 2r \sin \theta \sin \theta, \\ &= 2r(\cos^2 \theta + \sin^2 \theta) + \cos \theta = 2r + \cos \theta. \end{aligned}$$

$$\begin{aligned}
\frac{\partial g}{\partial \theta} &= \frac{\partial f}{\partial x} \frac{\partial x}{\partial \theta} + \frac{\partial f}{\partial y} \frac{\partial y}{\partial \theta} \\
&= (2x + 1)(-r \sin \theta) + (2y)(r \cos \theta) \\
&= (2r \cos \theta + 1)(-r \sin \theta) + 2r \sin \theta r \cos \theta \\
&= -2r^2 \sin \theta \cos \theta - r \sin \theta + 2r^2 \sin \theta \cos \theta \\
&= -r \sin \theta. \quad \square
\end{aligned}$$

Check. We check these results by noting that

$$g(r, \theta) = f(x(r, \theta), y(r, \theta)) = r^2 + r \cos \theta,$$

then by direct differentiation we find that

$$\frac{\partial g}{\partial r} = 2r + \cos \theta, \quad \frac{\partial g}{\partial \theta} = -r \sin \theta, \quad \text{as before.}$$

Example 3.8.3

Given $g(u, v) = f(x, y)$, $f(x, y) = x^2y$, $x(u, v) = uv$, $y(u, v) = \frac{u}{v}$, find g_u and g_v .

Solution.

$$\begin{aligned}\frac{\partial g}{\partial u} &= \frac{\partial f}{\partial x} \frac{\partial x}{\partial u} + \frac{\partial f}{\partial y} \frac{\partial y}{\partial u} \\ &= (2xy)v + (x^2)\frac{1}{v} \\ &= 2uv\frac{u}{v} + (uv)^2\frac{1}{v} \\ &= 2u^2v + u^2v = 3u^2v.\end{aligned}$$

$$\begin{aligned}\frac{\partial g}{\partial v} &= \frac{\partial f}{\partial x} \frac{\partial x}{\partial v} + \frac{\partial f}{\partial y} \frac{\partial y}{\partial v} \\ &= 2uv\frac{u}{v} - u^2v^2\frac{u}{v^2} = 2u^3 - u^3 = u^3.\end{aligned}$$

3.8.4 WARNING

For functions of one variable, it **is** true that $\frac{dy}{dx} = 1 \bigg/ \left(\frac{dx}{dy} \right)$.

$$\text{e.g. } y = \ln x \Rightarrow \frac{dy}{dx} = \frac{1}{x}.$$

$$\text{and } x = e^y \Rightarrow \frac{dx}{dy} = e^y = x = 1 \bigg/ \left(\frac{dy}{dx} \right).$$

However, this result is **not** true for partial derivatives.

Remark

Actually, this can be corrected with more care in notation and consideration of how the variables are fixed/dependent on each other. However, the moral here is that one has to be much more careful when dealing with partial derivatives and a more naïve approach may cause errors.

e.g. Plane polar coordinates

$$x = r \cos \theta, \quad y = r \sin \theta.$$

Then

$$\frac{\partial x}{\partial \theta} = -r \sin \theta.$$

However,

$$r(x, y) = \left(x^2 + y^2\right)^{\frac{1}{2}}, \theta(x, y) = \tan^{-1} \left(\frac{y}{x}\right).$$

Hence we see that

$$\frac{\partial \theta}{\partial x} = \frac{-y}{(x^2 + y^2)} = \frac{-r \sin \theta}{r^2} = \frac{-\sin \theta}{r}.$$

$$\text{i.e. } \frac{\partial x}{\partial \theta} \neq 1 \bigg/ \left(\frac{\partial \theta}{\partial x}\right).$$

3.9 Integration of PDEs

In a later part of the module we shall need to 'integrate' partial derivatives.

For example, given the ordinary differential equation $\frac{df}{dx} = 1$.

We know how to integrate this, and obtain $f(x) = x + c$, where c is the 'constant of integration'.

But suppose now that f is a function of two variables x and y , so that $f = f(x, y)$. Suppose also that we were given the *partial* differential equation

$$\frac{\partial f}{\partial x} = 1.$$

What is the solution this time?

3.9 Integration of PDEs (ctd)

Given $\frac{\partial f}{\partial x} = 1$, clearly, $f(x, y) = x + c$ is still a solution, where c is a constant. But is this the most general form of the solution?

Recall that differentiating partially with respect to x treats y as a constant (and vice versa). So in the solution we have just obtained, the constant c might depend upon y and nothing will change.

Indeed, if we write

$$f(x, y) = x + g(y),$$

then

$$\frac{\partial f}{\partial x} = \frac{\partial x}{\partial x} + \frac{\partial g(y)}{\partial x} = 1 + 0 = 1.$$

So when integrating partial derivatives, the ‘constant’ of integration will generally depend upon the *other* variable.